

Networks Sub-module Assignment

Answers for Part 2 and Part 3

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Part 2:

1. Which of these frames will be received by selected ports:

Port 0: Frame 1 and Frame 2

Port 1: Frame 0, Frame 1 and Frame 3

Port 2: Frame 0, Frame 1 and Frame 2

Port 3: Frame 0 and Frame 2

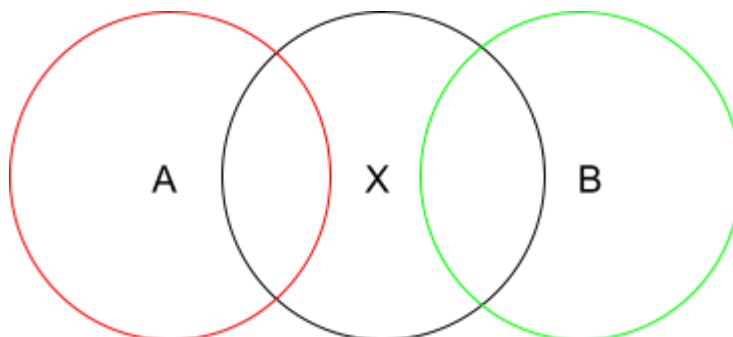
2. What is the switching table after forwarding these 4 frames:

Port	Mac Address
0	60-4A-18-B2-63-DA
1	10-0C-2B-AF-18-7B
2	
3	AC-69-D6-57-24-A3

Port 2 is not filled as Device 2 didn't send a frame so the switch table doesn't add it.

Part 3

- 1.



2. Analyse the above transmission situation and describe the transmission procedure.

A starts at 120 and finishes at 320.

B starts at 340 and finishes at 440

Time(μ s)	Events
0	X is sending packets and will need to take 100 μ s
30	A wants to send a packet that requires 200 μ s but needs X to stop transmitting first
70	B wants to send a packet that requires 100 μ s but needs X to stop transmitting first
100	Now X has stopped both A and B can begin to wait for their DIFS timers to count down
110	Now that the DIFS time has expired both A and B start counting down their backoff timers respectively
120	A's backoff timer is lower than B's (10 μ s vs 20 μ s) so A begins sending its frame to X. B's backoff timer doesn't reset however it just pauses so now B's backoff timer is 10 μ s
320	A has finished transmitting and now the DIFS timer can begin counting down
330	As mentioned before B's backoff counter now starts at 10 μ s rather than 20 μ s and begins counting down
340	B begins transmitting its frame to X
440	B finishes transmitting its frame to X